Samuel Martineau

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Skills

- Native French speaker; perfectly fluent in English
- Experienced in a wide variety of programming languages and paradigms
  - Strongest in: JavaScript / TypeScript, Zig, Swift, Python, etc.
  - Familiar with: C, Haskell, C++, Fortran, Java, etc.
- Experienced in Unix system administration (web servers, databases, services, containers, etc.)
- Comfortable in Computer Aided Design and additive manufacturing
- Experienced Nix user (from development environments, to infrastructure management) and package maintainer

Education

University of Waterloo *Bachelor of Science* — Fall 2024 — Spring 2029
Mathematical Physics, Honours, Co-operative Program with Minor in Pure Mathematics — 95.29% CGPA

Experience

University of Waterloo *Undergraduate Researcher* — Summer 2026 (Ongoing)
Working with Dr. David Yevick on the application of modern machine learning techniques to ultrasound imaging of the mitral valve, in collaboration with MIT

INKAS Industries *Applied Science Researcher* — Winter 2026
Helped kickstart a new drone research and development department, at the intersection of software and hardware ⋆ Led hardware procurement ⋆ Designed project plans based on governmental agency requirements ⋆ Collaborated on software connecting modern artificial intelligence with verifiable reasoning

Solana Foundation *Software Developer (independent contractor)* — Fall 2025
Led a team of four through the development of the first feature-complete software development kit for the Solana Blockchain on Swift ⋆ Ensured idiomatic design and alignment with core tenets of iOS platform ⋆ Developed a variety of low-level libraries for fast interaction with the blockchain

Wolfram Research *Student Ambassador (volunteer)* — 2023 — Present
Built a variety of technical demonstrations of the Wolfram platform, mainly Mathematica, in multiple fields, including visualizations of chemical and physical phenomena targeted at high school students ⋆ Introduced Mathematica, and provided technical support, to peers in local community

Xata *Software Developer (intern)* — Summer 2024
Built internal tooling for customer information search and retrieval ⋆ Reviewed and audited existing code, and provided improvements to the linting system ⋆ Identified and diagnosed a critical performance issue in core product

World Affairs Conference *Director of technology (volunteer)* — 2023 –2024
"North America's largest and Canada's oldest annual student-run current events conference" ⋆ Built internal tooling for outreach and attendee management ⋆ Created a content management system ⋆ Designed an algorithm for attendee-plenary-room assignments based on Gale and Shapley's solution to the stable marriage problem

Camps Modulo *Scientific animator* — Summer 2022
Developed a variety of safe and entertaining projects demonstrating key principles of physics, as well as a variety of chemistry demonstrations ⋆ Worked with kids ranging from 5 to 12 years old

Projects

Experiments in Electricity & Magnetism Currently investigating the topological impact of position on voltmeter measurements around an induction coil, with the goal of developing an interesting demonstration of the weirdness of electromagnetism to first-year undergraduate students.

Various science fairs Analysis of impact of various solvents on chewing-gum (2017—2018), Creation and testing of a pedagogical video game to practice 1st grade mathematics (2018—2019), Development of interactive simulator to study natural evolution (2019—2021), Invention and prototyping of a wall-climbing robot (2021—2022) ⋆ Won variety of awards: First Hydro-Québec Prize, Award of the Canadian Meteorological and Oceanographic Society, Québec Order of Chemists Award, Francis Boulva Recognition Award, etc. ⋆ Multiple appearances on national television: Génial! (2019, 2022, 2023) and Les inventifs (2023)

SolShare (Hack the North 2025)

- Built a native iOS application in SwiftUI to allow roommates to easily share bills and track expenses
- Utilizes multimodal LLMs with self-critiquing to reliably itemize receipts from pictures
- Relies on the Solana blockchain to allow for instant and easy settlement of debts
- Won over $8000 in prizes with the Best Use of Solana and Best Use of Cohere awards